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The morphology-dependent on the electrochemical performance of molybdenum trioxide in supercapacitors

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ABSTRACT

The energy storage performance of transition metal oxides (TMO) as electrode materials for supercapacitors exhibit a strongly morphology-dependent due to changes in the local electronic and microscopic geometric structure of the interface. Herein, MoO_3 with morphology in nanoprisms (NP), nanostrips (NS), and nanobulks (NB) were prepared by different synthesis routes. The morphology and structural characteristics of the three obtained MoO_3 were characterized by scanning electron microscopy, the high resolution transmission electron microscopy, Brunauer–Emmett–Teller techniques, X-ray diffraction, and X-ray photoelectron spectroscopy. And then, the significant effects of the morphologies of MoO_3 on the electrochemical activity were assessed using cyclic voltammetry and galvanostatic charge-discharge measurements. The supercapacitors using MoO_3-NP as electrode achieved the largest specific capacitance of 254.0 F g^{-1} at a current density of 0.3 A g^{-1} as compared to 123.3 F g^{-1} of MoO_3-NS, and 8.7 F g^{-1} of MoO_3-NB in 1 mol L^{-1} Na_2SO_4 electrolyte. Among them, the MoO_3-NP electrode has the highest pseudo-supercapacitive response and excellent cycling stability than the other electrodes, which is attributed to the morphology-dependent to enhance the electrical conductivity and promote the generation of active reaction sites. Therefore, the MoO_3-NP electrode can be significantly applied to supercapacitors for cost reduction and energy conversion efficiency improvement.

1. Introduction

Supercapacitors are a class of well-established of energy storage device that can store electrochemical energy directly between traditional batteries and capacitors, and have the advantages of high power and energy density, fast charge/discharge speed, long cycle stability, green environment and no memory effect, which can play a vital role in supplementing the energy storage functions of batteries by providing back-up power in the event of a blackout [1,2], and are now widely used in portable electronics, medical devices, power supply, hybrid vehicles and other fields [3,4]. Supercapacitor are classified as electric double-layer capacitors (EDLCs) and pseudocapacitors (PsCs) [5,6]. EDLCs use a polarized electrode/electrolyte interface to form a bilayer of the charge separation of electrons and ions to achieve electrostatic energy storage based on physical adsorption/desorption without charge transfer. The electrode materials are mainly made of carbon materials with large specific surface area and high activity [7,8]. PsCs undergo redox reactions at the characteristic potential of the electrode/electrolyte interface to generate Faraday quasi-capacitance [9,10]. The electrode materials of PsCs mainly include transition metal compounds (TMCs) [11–13], conductive polymer [14,15] and multi-component compounds [16–18], etc. Among them, the research and synthesis of transition metal oxides (TMOs) as a type of TMCs has received great attention in advancing energy storage to increase energy and power density [19,20]. The noble RuO_2 as TMOs was commonly used as electrode materials of PsCs [21,22]. In particular, RuO_2 using sulfuric acid solution as the electrolyte can generate high current density by combining the resulting double-layer process [23,24]. However, the low abundance and high price of Ru make it difficult to be commercialization [25]. To solve this issue, the selection of other TMOs as substitutes for RuO_2 as PsCs electrode materials, while maintaining the high power and long life, has achieved tremendous attention in the field of supercapacitor devices. Therefore, A variety of pseudocapacitive TMOs materials have been prepared, such as MnO_2 [26], NiO [27],

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V_2O_5, VO_2, V_2O_3 [28,29], CuO [30], Co_3O_4 [31], Nb_2O_5 [32], CeO_2 [33], MoO_2, MoO_3 [34,35], Fe_3O_4 [36], MnO_2 [37], Mn_3O_4 [38], etc. Although some progress has been made, the selection of a suitable TMO electrode material of PsCs that is efficient, stable, low cost and readily accessible remains an important research focus for improving the electrical storage performance of supercapacitors.

PsCs energy storage are highly reversible faradic redox reactions or adsorption/desorption processes that occur on the two-dimensional surface or body phase quasi two-dimensional space of electrode materials, which utilize fast electrochemical reactions to achieve charge storage [39]. Even TMOs of the same composition have different morphologies, resulting in different microelectronic and geometrical structures, which directly affect the adsorption-desorption and activation state of the reactant molecules at the reactive interfacial surfaces, thus exhibiting different reactivity [40,41]. Viswanathan et al. used radio-frequency magnetron sputtering to coat 2D α-MoO_3 nanosheets onto carbon cloth. In 1.0 mol L^{-1} aqueous Na_2SO_4 electrolyte, the specific capacitance was 240 F g^{-1} at a current density of 1.5 mA cm^{-2}, and the capacity was maintained at 78.8 % after 5000 cycles at 2.5 mA cm^{-2} [34]. Qi et al. synthesized the porous Mn_2O_3 using a hard template method. At discharge current density of 0.2 A g^{-1}, the specific capacitance of the porous Mn_2O_3 was 102 F g^{-1}, higher than that of 4.88 F g^{-1} and 2.28 F g^{-1} of the two bulk Mn_2O_3 [43]. Joo et al. evaluated the effect of different metal precursors (acetate, nitrate and chloride) on the morphology-dependent of fabricated $MnCo_2S_4$ on nickel foam by in situ hydrothermal method on supercapacitor performance [42]. As a result, in the design process of supercapacitors, the electrode material is a key issue that affects the electrochemical performance of supercapacitors, which directly affects the properties of electrochemical reactions such as potential and conductivity. The ideal electrode material should have a high specific surface area, excellent electrical conductivity and stability. In particular, for the selected electrode materials of TMOs, the specific surface area and pore structure based on morphology are two important factors to determine their electrochemical properties. Thus, solving the morphology-dependent of TMOs electrode materials become one of the very interesting research directions for realizing PsCs with ideal power and energy density.

Among TMOs, MoO_3 as potential TMOs electrode material in PsCs possesses distinct advantages in terms of a simple preparation, superior thermal and mechanical stability, large theoretical specific capacitance, low cost and eco-friendly, which has received widespread research attention due to high power and energy density [44]. Generally, the MoO_3 exist in three different crystalline polymorphs in the thermodynamically metastable hexagonal phase (*h*-MoO_3), the monoclinic phase (β-MoO_3), and the stable orthorhombic phase (α-MoO_3) [45]. Remarkably, Mo-based oxides exhibit multiple redox valences from +2 to +6 in different redox reactions, enhancing the high electrochemical activity. Additionally, the MoO_3 provides a wider operating voltage window for PsCs due to its high work function [46]. Based on the above considerations, three types of different morphologies of MoO_3 with nanoprisms (NP), nanostrips (NS), and nanobulks (NB) were prepared by different methods. In this work, the research mainly focused on the morphology-dependent of the MoO_3 electrode, and the capacitive properties of the MoO_3 electrodes were compared in parallel. Extensive and in-depth studies have illustrated that different morphologies of MoO_3 change the surface electronic and geometric properties. As a result of the corresponding various electrochemical characterization, the mesoporous structure MoO_3-NP can provide a larger surface area for adsorption of ions and more accessible channels for diffusion of ions, both of which are beneficial for the faradic redox reaction to achieve significant capacity performance in PsCs.

2. Experimental

2.1. Chemicals

Na_2SO_4 and HNO_3 (purity≥99 %) were supplied from Tianjin Chemical Reagent Company. $(NH_4)_6Mo_7O_{24}·4H_2O$ (molecular weight: 1235.86, purity ≥99.0 %) and anhydrous ethanol were purchased from Beijing Chemical Reagent Company. Carbon cloth was received from WOS 1002 PHYCHEMi Co. Ltd., China. The chemical reagents used are analytical grade purchased directly and not purified in this work.

2.2. MoO_3 electrodes preparation

The carbon cloth was cut into substrates of 1.0 × 1.5 cm^2 as substrate, thoroughly washed in an anhydrous ethanol ultrasonic cleaner for 10 min to remove impurities, baked in a drying oven for 30 min, and accurately weighed the mass of each piece of carbon cloth to be about 18 mg cm^{-2}. Three morphologies of MoO_3 use different preparation methods.

(1) MoO_3 nanoprisms (NP): 1.5 g of $(NH_4)_6Mo_7O_{24}·4H_2O$ was dissolved in 50 mL of deionized water with stirring for 10 min to obtain a colourless and transparent solution, then 2.2 mol L^{-1} HNO_3 was dropwise added to the above solution until the pH was adjusted to the desired value of 1.5 with continuous stirring for 1 h. The mixed yellow solution was then poured into a 50 mL of Teflon-lined stainless steel autoclave, while several well-cleaned strips of carbon cloth as the substrate of PsCs electrode were put into the reactor and pressed vertically against the Teflon liner. The autoclave was sealed and kept in an oven and gradually heated from 25 to 180 °C in a thermal rate of 2 °C·min^{-1}, and the reaction continued for 24 h.

(2) MoO_3 nanostrips (NS): The preparation of MoO_3-NB is the same as the first half of the preparation process (1) for MoO_3-NP, except that the oven temperature was directly maintained to 180 °C, and then the autoclave was placed in the oven for 48 h without a step-by-step heating process.

In (1) and (2), each mixture solution was reacted by a simple hydrothermal method for the appropriate time, and the autoclave was gradually lowered to room temperature. The MoO_3-NP and MoO_3-NS nanocrystals had been already grown onto the carbon cloth substrate, then washed several times with anhydrous ethanol and deionized water to remove residual ions, and dried in a vacuum oven at 60 °C for 30 min to remove residual adsorbed gas.

(3) MoO_3 nanobulks (NB): 2.0 g of $(NH_4)_6Mo_7O_{24}·4H_2O$ was directly pyrolyzed using a high-temperature vacuum tube furnace with controlled temperatures from 150 to 650 °C for 3 h at a thermal rate of 10 °C·min^{-1} under air flow rate (50 sccm), and the optimized pyrolysis temperature is at 500 °C. The residual gas portion of the decomposition and volatilization of $(NH_4)_6Mo_7O_{24}·4H_2O$ was carried by the air flow and absorbed into the tail gas bottle, which can avoid the uptake and contamination of the target MoO_3.

The MoO_3-NP and MoO_3-NS were prefabricated on carbon cloth strips during the reaction process, and directly used as electrodes for supercapacitors. The obtained light green MoO_3-NB particles were dispersed with 4 mL of N-methylpyrrolidone by ball milling, and mixed with the adhesive PVDF. And then, the as-paste was coated on the carbon cloth, which was dried in a N_2 atmosphere at 60 °C for 30 min. The MoO_3-NB electrode was used for supercapacitors. The mass of the different MoO_3 loaded on the electrode was selected and measured by using gravimetric method to be about 2 mg according to different preparation methods. The electrochemical measurements were

performed using a two-compartment glass cell with a three-electrode configuration in the aqueous electrolyte 1.0 mol L^{-1} Na_2SO_4 solution at room temperature, which were measured using the as-fabricated three different morphology of MoO_3 electrode, platinum foil (0.5 × 2 cm^2), and a saturated calomel electrode (SCE), as the working electrode, counter electrode, and reference electrode, respectively.

2.3. Measurements

The surface morphology of three MoO_3 electrodes was examined using scanning electron microscopy (SEM, Hitachi, S-4800, Japan) at a 15 kV acceleration voltage. The high-resolution transmission electron microscopy (HETEM) attached with the selected area electron diffraction (SAED) is performed on the ESCALAB250 (Thermo VG, USA). The X-ray diffraction (XRD) is collected with the D8 Bruker powder X-ray diffractometer (Cu-Kα 0.1541 nm, Ni filter) to determine the crystal structure at $1^{\circ}\cdot s^{-1}$ in the range 10–80°. X-ray photoelectron spectroscopy (XPS) is performed by Thermo Scientific K-Alpha with Al Kα $h\nu$ = 1486.6 eV at an voltage of 12.0 kV and an emission current of 6 mA and at a vacuum of $\approx 10^{-9}$ mbar during spectra acquisition. In particular, according to the relevant literature [47,48], due to the wide variation of C 1s peak positions in adventitious carbon (AdC), which may lead to unphysical results for linear correction of the binding energy (BE) scale, all the core-level spectra of the XPS analysis were calibrated by the sample work function. The Brunauer–Emmett–Teller (BET) is determined using an Antosorb-1 apparatus system (Quantachrome, USA) with nitrogen as adsorbent. Cyclic voltammetry (CV), galvanostatic charge/discharge characteristics (GCD), and Electrochemical Impedance Spectroscopy (EIS) are carried out with a CHI 660C workstation (Chenhua Shanghai). EIS is performed within the frequency range between 0.1 Hz and 100 kHz with alternate current amplitude of 10 mV at open circuit potential. The cycle life of the galvanostatic charge/discharge was examined by the blue battery (LAND, CT2001A, Wuhan).

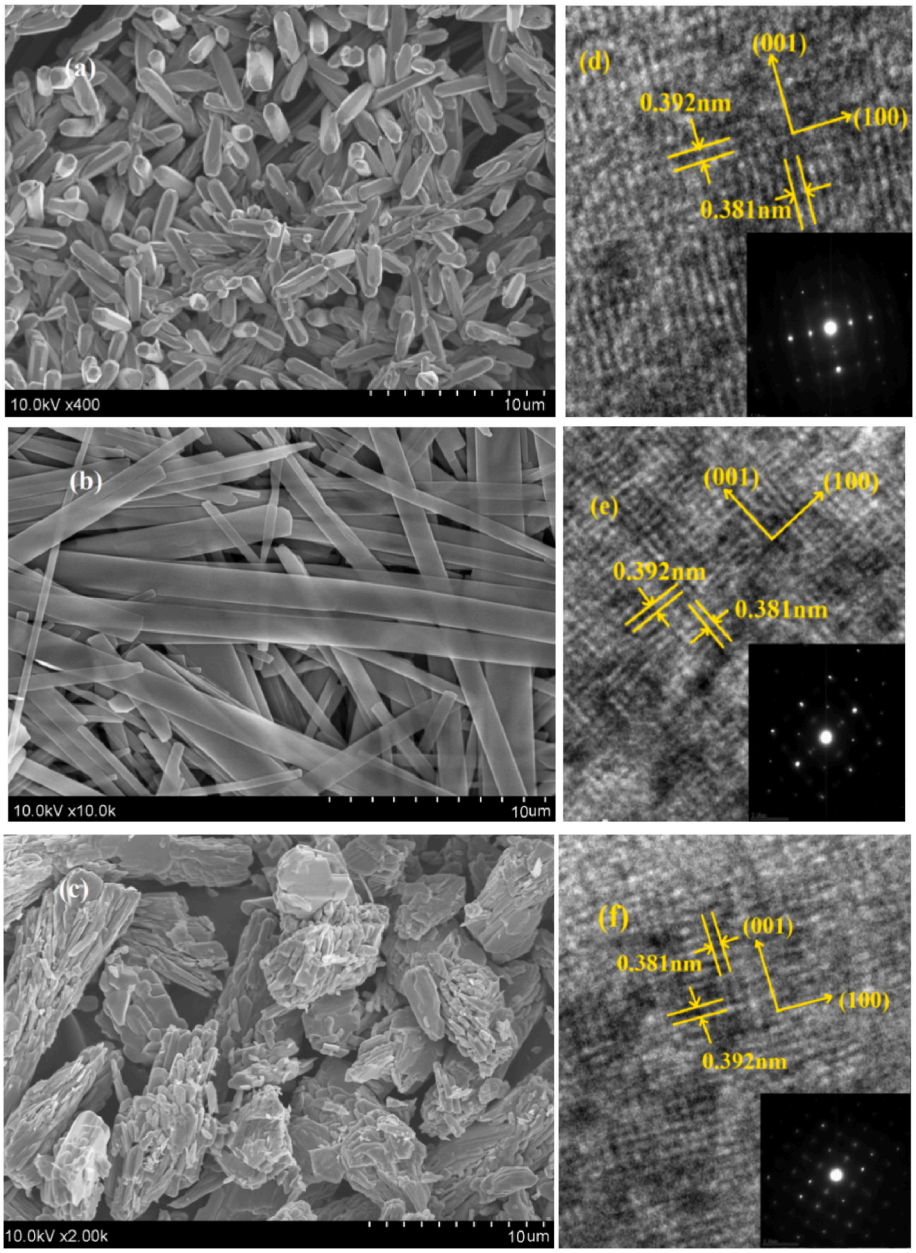


Fig. 1. Sem of (a) MoO_3-NP, (b) MoO_3-NS, (c) MoO_3-NB and HR-TEM of (d) MoO_3-NP, (e) MoO_3-NS, (f) MoO_3-NB.

3. Results and discussion

3.1. Characterization of MoO_3 electrodes

The SEM and HR-TEM of three different of MoO_3 were investigated in Fig. 1(a–f), all of which are MoO_3 with different crystallographic orientations and shapes. As can be seen from Fig. 1(a), MoO_3-NP exhibits a flower-like structure with octahedral prism particles clustered together, and with a lateral diameter of about 10 nm and a length of about 40 nm, which had a smooth surface and fairly uniform distribution. It is clear from Fig. 1(b) that MoO_3-NS presents a 2D rectangular long strip-like morphology without pores, and with a very thin thickness of about 2–3 nm and unequal width and length, which were designated as nanostrips rather than other nanostructures. Fig. 1(c) displays that the typical agglomerated MoO_3-NB is made from nanocrystals consisting of regular stacks of Mo-octahedrons with irregular in-plane ordering, thus, presenting an apparently consolidated morphology with bulk shape and rough surface. The preparation methods make a measurable effect on the microstructure of the different morphologies of MoO_3, which were further characterized by HRTEM in Fig. 1(d–f) and the SAED inset in Fig. 1(d–f). The typical HRTEM images show that MoO_3-NP, MoO_3-NS, and MoO_3-NB all have highly ordered lattice structures with clear lattice fringes and periodic diffraction arrays. The two sets of parallel fringes in two directions at 90° angle to each other with spacing of 0.392 nm and 0.381 nm, corresponding to the (100) and (001) planes of the MoO_3, respectively [49,50]. The indexed SAED spots of MoO_3-NP can be attributed to the (110), (201), (041), and (002) planes with principal growth direction along the (001) and the lateral growth along the (100) and (010) directions, which might be undergo the self-aggregation or continue to grow through the elongation along 1D, leading to the highly anisotropic crystalline growth of α-MoO_3 thin nanoprisms [50–52]. And the SAED pattern of MoO_3-NS and MoO_3-NB recorded primarily perpendicular to the long side corresponds to (001) plane, and the growth axis was identified as to the (010) zone axis, indicating non-synchronous oriented growth along the (001) direction with exposed top/bottom facets of (010) [53]. SAED pattern reconfirms that the MoO_3 are characterized with excellent material crystallinity.

The textural characteristics of MoO_3-NP, MoO_3-NS, and MoO_3-NB were respectively measured by the corresponding Brunauer–Emmett–-Teller analyzes. All isotherms in Fig. 2(a) are identified as type IV with a H4 hysteresis loops in the N_2 adsorption/desorption branch at P/P_0 = 0.0–1.0, indicating mesopores materals [54]. According to the calculation results of an assumption of slit pore model of the non-local density functional theory (NLDFT) method, three kinds of MoO_3-NP, NS and NB presented the BET specific surface area (S_{BET}) values of 51.23, 18.16, 5.01 $m^2\ g^{-1}$, the pore volume (V_p) values of 0.058, 0.028, 0.007 $cm^3\ g^{-1}$ and the average pore diameter distributions (d_p) *of* 3.36, 3.04, 3.35 nm, *respectively*. The nitrogen adsorption-desorption investigation was in good agreement with the mesoporous structure of three different morphology MoO_3 revealed by SEM. Furthermore, in order to compare the conductivity, the square resistance of MoO_3-NP, MoO_3-NS, and MoO_3-NB were of 1.02 S m^{-1}, 1.47 S m^{-1} and 1.65 S m^{-1}, respectively. There was no significant difference in conductivity between three different morphologies of MoO_3.

Fig. 3(a) presents the influence of the synthetic method on the phase state, crystallite of the MoO_3 with different morphology (MoO_3-NP, MoO_3-NS, and MoO_3-NB). The XRD diffraction peaks of MoO_3-NP at 2θ of 12.80°, 23.35°, 25.73°, 27.34°, 33.77°, 35.13°, 38.98° and 49.27° match well with the (020), (110), (040), (021), (111), (041), (060), and (002) lattices of the standard orthotropic phase α-MoO_3 (JCPDS: 50–508) without other impurity peaks. The MoO_3-NS and MoO_3-NB presented an orthorhombic phase together with a minor percentage of hexagonal phases. The corresponding planes (110), (200), (210), (300), (310), (320) and (410) exhibited at 2θ values of 16.75°, 19.45°, 25.73°, 29.36°, 35.13°, 43.10° and 45.47° belong to the standard hexagonal *h*-MoO_3 (JCPDS: 21–569) impurity phase [55]. The three morphologies of MoO_3 are preferentially oriented in different planes, but all have good crystallinity, with only diffraction peaks of different intensities. Moreover, the surface redox valence states in the different morphologies of MoO_3 were determined by XPS. The XPS survey spectra of the three MoO_3 in Fig. 3(b) all presented four mainly high-intensity peaks, which can be attributable to Mo 3d, Mo $3p_{3/2}$, Mo $3p_{1/2}$, and O 1s. The peak at 973.65 eV (O-*KLL*) represents the energy emitted due to the filling of the O 1s state (*K* shell) by an electron from the *L* shell coupled with the ejection of an electron from the *L* shell. It is worth noting that recent literatures reported that aligning the spectrum to the C 1s peak of the conventional AdC as reference is unreliable because the C–C/C–H peak can be arbitrarily set to any value between 284.0 and 285.6 eV depending on the substrate [56,57]. Therefore, according to the literature [58], the work function (Φ) of the tested sample can be used to correct the C 1s peak via the relation E_B = 289.58 eV – Φ for the calibration of the core-level spectra of XPS analysis [59]. The Φ of the sample (MoO_3) is 6.03 eV based on the ultraviolet photoelectron spectroscopy (UPS) tests as shown in Fig. 3(g), in which the onset (E_{onset}) and cut-off (E_{cutoff}) energy regions are on the left and right respectively. The W_f of prepared MoO_3 are determined by the equation of W_f = 21.21 eV - (E_{cutoff} - E_{onset}). Hence, the binding energy of C 1s peak of AdC is set at 283.55 eV and all core level scans are corrected accordingly. Fig. 3(c–e) show the XPS spectrum in the Mo 3d binding energy region for an as-prepared MoO_3-NP, MoO_3-NS, and MoO_3-NB with two well-fitted spectral peaks identified as the characteristic Mo 3d spin-orbit doublet peaks for Mo $3d_{5/2}$ and for Mo $3d_{3/2}$, which can be deconvoluted into four parts by fitting. The existence of two major peaks at 230.9 eV and 234.1 eV are ascribed to the oxidation state of Mo^{6+}. The comparatively small peaks at about 232.5 eV and 229.4 eV indicate the existence of the Mo^{5+} oxidation state in MoO_3. The existence of Mo^{5+} oxidation state is

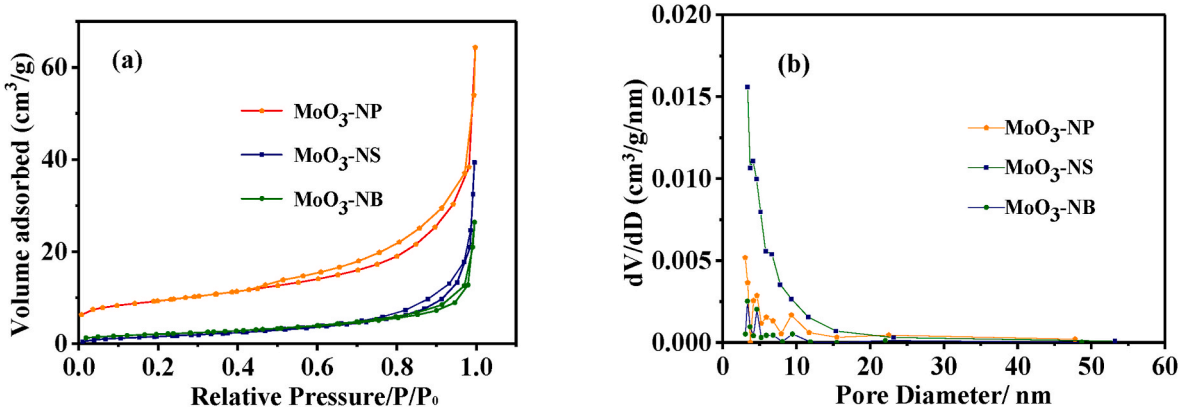


Fig. 2. (a) BET isotherms and (b) d_p of three different morphology of MoO_3.

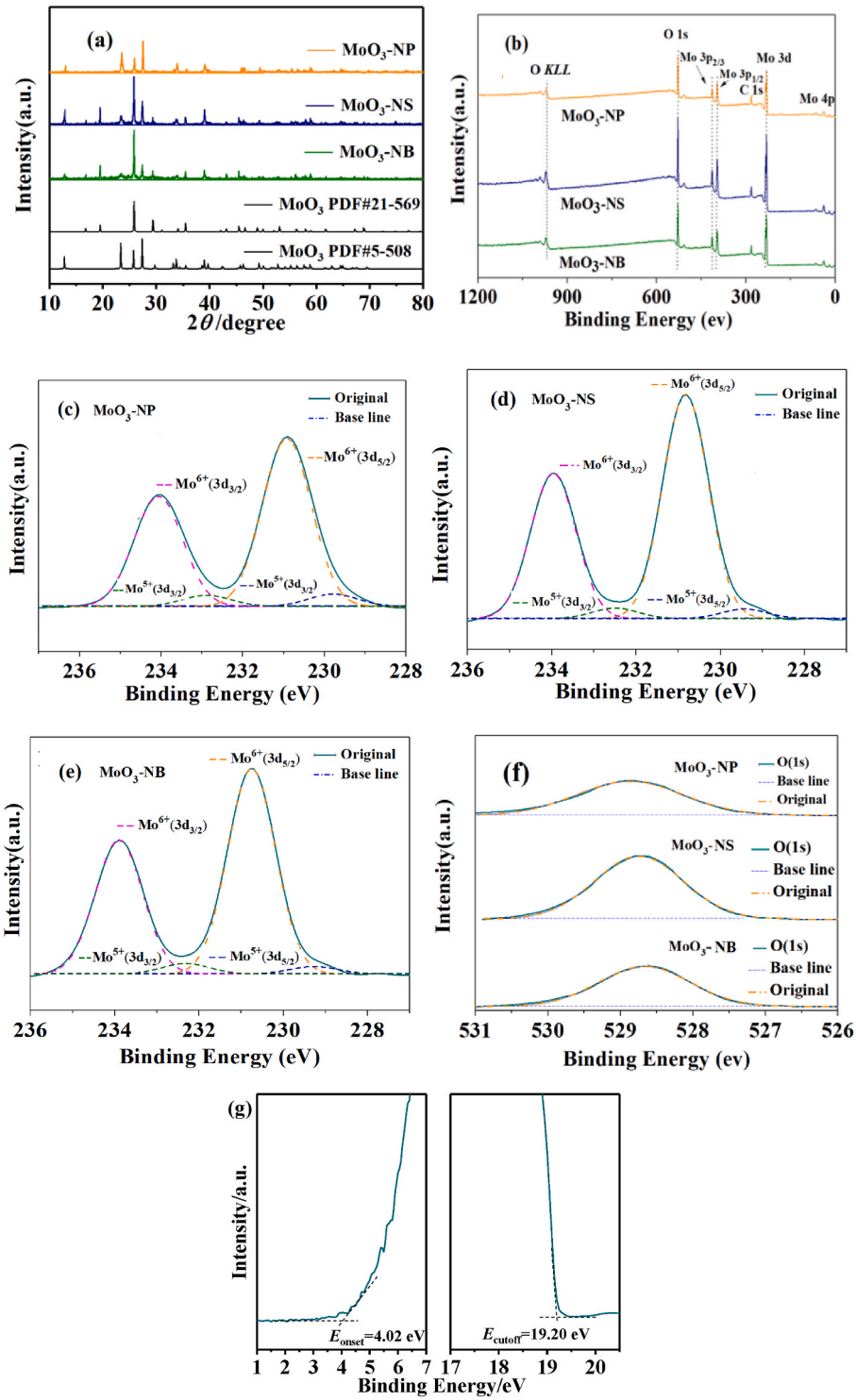


Fig. 3. (a)XRD, (b–f) XPS of three different morphology of MoO_3, (g) UPS spectra of MoO_3.

related to the oxygen vacancy defect in the lattice of MoO_3 [54]. The intense characteristic peak of O 1s was shown around 528.5 eV in Fig. 3 (d), and may have originated from the chemical bonds of Mo-O in the lattice of MoO_3 [44]. The SEM, HR-TEM, XRD, and XPS measurements proved that the different morphologies of MoO_3 were synthesized.

3.2. The electrochemical performance of the different MoO_3

Fig. 4(a–c) show the cyclic voltammetry (CV) profiles of the different morphology MoO_3 electrodes performed between the potential range −0.2 to 0.5V with a three-electrode configuration at different scan rates of 20, 40, 60, 80 and 100 mV s^{-1} in 1 mol L^{-1} Na_2SO_4 electrolyte to characterize the electrochemical activities. The CV curves of the MoO_3 electrode with different scan rate illustrates that the reduction/oxidation peaks currents are linearly increased with an increase in the scan rate. Here, all CV profiles of the different morphologies of MoO_3 were attributed to the typical characteristic pseudo-capacitive charge storage [60]. Among these three electrodes, a more diffusion controlled process was observed in the MoO_3-NS and MoO_3-NB electrode of supercapacitor. Hence, the rate of the diffusion-controlled reaction to reduce the

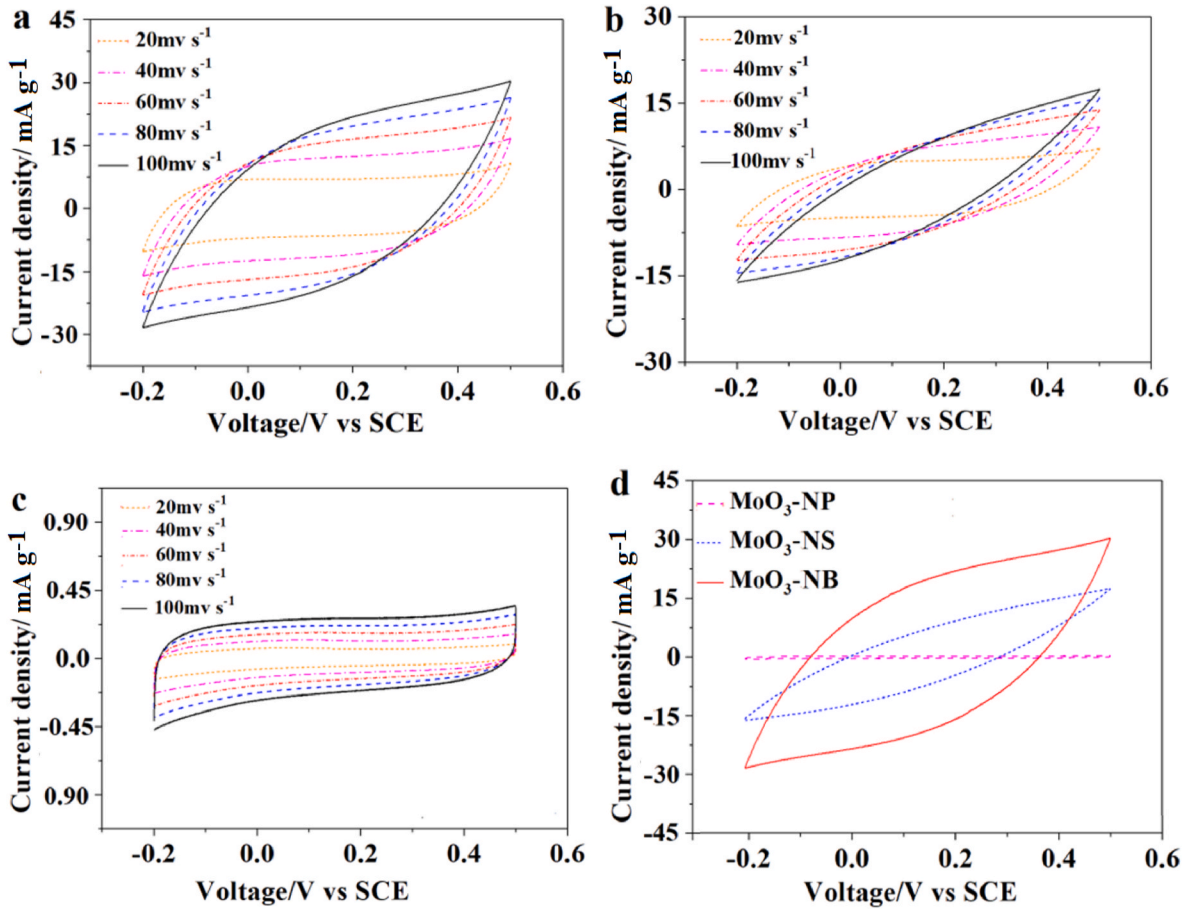


Fig. 4. CV profiles of (a) MoO_3-NP, (b) MoO_3-NS (c) MoO_3-NB as electrode at different scan rates, and (d) the different morphology MoO_3 at a scan rate 100 mV s^{-1}.

supercapacitor storage cyclic and rate capability is limited. But, the pore-rich more porous nature of the MoO_3-NP showed a higher percentage of capacitive contribution than that of the less porous MoO_3-NS and MoO_3-NB [61]. The representation of electrochemical Na^+ insertion mechanism can be expressed as the following eqn. (1):

$$MoO_3 + xNa^+ + xe^- \leftrightarrow Na_xMoO_3 \quad (1)$$

The mechanism involves the intercalation of Na^+ into the electrode

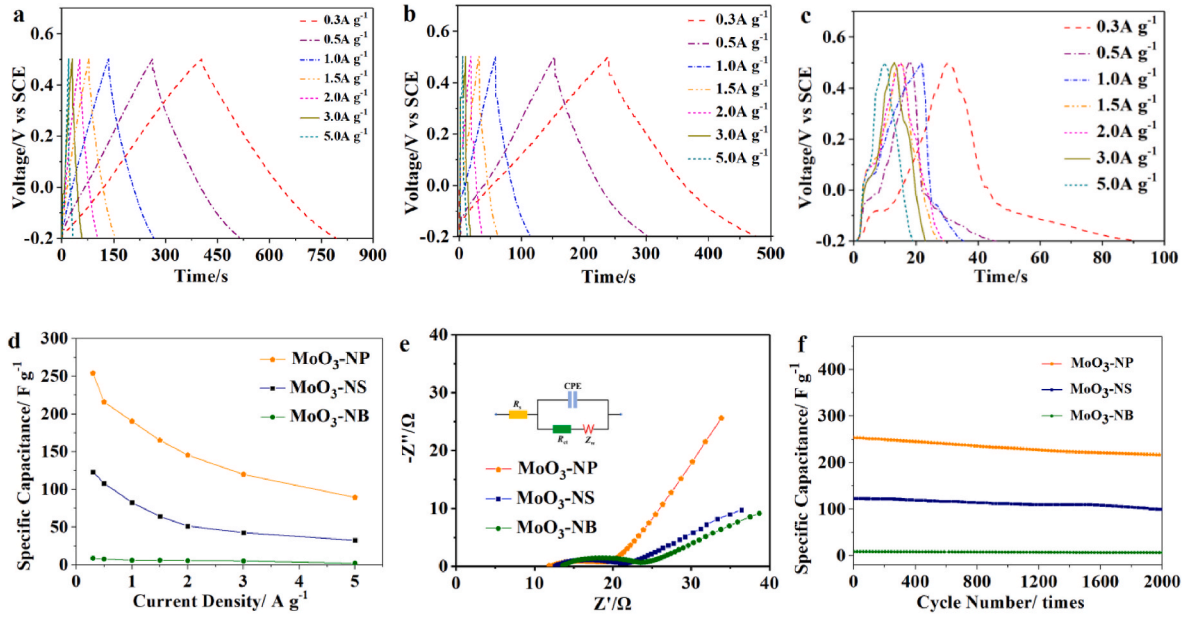


Fig. 5. GCD curves of (a) MoO_3-NP, (b) MoO_3-NS, (c) MoO_3-NB electrodes at different current density, (d) Specific capacitance of the different morphology MoO_3, (e) Nyquist plots of the different morphology MoO_3 (f) the long-term cycle stability during 2000 cycles of three different morphology MoO_3 electrodes at 0.3 A g^{-1}.

material during reduction and de-intercalation upon oxidation [62]. Compared with the different morphology MoO_3 electrode, the redox currents of MoO_3-NP electrode were larger than that of MoO_3-NS and -NR electrodes, and the CV profiles were almost close to rectangular-like shapes with good symmetry and larger integral area, implying good charge transfer performance and capacitance characteristics. At the same scanning rate, the response currents of MoO_3-NS and -NB electrodes are significantly lower than those of MoO_3-NP, as shown in Fig. 4 (d). In addition, with the increase of the scanning rate from 20 to 100 mV s^{-1}, the current densities of the different MoO_3 increase in the same time, and there was no noticeable distortion in the CV profile, which indicated a stable pseudo-capacitance behavior and significant ionic response.

The galvanostatic charge-discharge (GCD) curves of the different morphology MoO_3 as electrodes in PsCs were performed with current densities of 0.3, 0.5, 1.0, 1.5, 2.0, 3.0 and 5.0 A g^{-1} in Fig. 5. It is clear that MoO_3-NP and MoO_3-NS electrodes display an approximately symmetric triangular shape, indicating that it is consistent with the classical double-layer capacitor charge/discharge, while MoO_3-NR exhibits significant bending. By comparing galvanostatic charging-discharging curves in Fig. 5(d), the MoO_3-NP electrode has the largest specific capacitance and the longest charge/discharge time at each current density. At 0.3 A g^{-1}, the MoO_3-NP, MoO_3-NS and MoO_3-NB were 254.0, 123.3, and 8.7 F g^{-1}, as well as the charge/discharge time were 794 s, 477 s and 91s, *respectively*. As a result, the specific capacitance of the different morphology MoO_3 electrodes depend on the charge/discharge current density, which gradually increase with the decreasing of current density due to the accessibility diffusion effect and ion migration at low current density, thus, the active sites on electrode are used with greater effectiveness. As the current density decreases from 5.0 A g^{-1} to 0.3 A g^{-1}, the specific capacitance values of the MoO_3-NP, MoO_3-NS and MoO_3-NB increased from 89.2 to 254.0 F g^{-1}, 32.6–123.3F·g^{-1} and 2.1–8.7 F g^{-1}, *respectively*. In contrast, the corresponding retention values were approximately 35.1 %, 26.5 % and 24.1 % of the initial specific capacity. Importantly, the specific capacitance value of MoO_3-NP electrode was superior to that of MoO_3-NS and MoO_3-NB at any current density. For example, the MoO_3-NP electrode had achieved a specific capacitance value of up to 254.0 F g^{-1} at 0.3 A g^{-1}, which was almost 2.1 times and 29.2 times that of the MoO_3-NS and MoO_3-NB electrodes, respectively. This result clearly demonstrates the morphological advantage of the MoO_3-NP with more active sites compared to other morphologies, indicating that the MoO_3-NP with the highest electrochemical performance was superior to the other two MoO_3 electrodes.

To further understand the electrochemical behavior of three different morphologies of MoO_3 as PsCs electrodes, EIS were measured to compare the charge transfer resistance (R_{ct}) in 1 mol L^{-1} Na_2SO_4 electrolyte. The Nyquist plots of the different morphology MoO_3 electrodes are similar in shape in Fig. 5(e). R_s values of the MoO_3-NP, MoO_3-NS and MoO_3-NB were 12.04, 12.63 and 13.61 Ω cm^2. R_{ct} values are 6.45, 8.80 and 10.31 Ω cm^2, *respectively*. The MoO_3-NP with the largest S_{BET}, high interconnectivity void space and porosity that not only generated the electrochemical active units, but also promote the adsorption and diffusion of reactive molecule for faradic behavior at the electrolyte/electrode interface, thus, resulting in the decrease of R_{ct}, which is in good concord with the results of S_{BET} of BET analysis [63]. Also, compared with the MoO_3-NS and MoO_3-NB, the straight lines of the MoO_3-NP showed a sharp slope at the low-frequency, which proved a rapid diffusion behavior with lower diffusion impedance and the superior capacitance [64]. Therefore, the EIS results further verify the morphology-dependent contribution to the charge diffusion behavior between the electrode and electrolyte interface to enhance the performance of supercapacitors.

The cycling stability of three different morphologies of MoO_3 as PsCs electrodes was tested over 2000 continuous charge/discharge cycles at 0.3 A g^{-1}. As shown in Fig. 5(f), the specific capacitance of the MoO_3-NP decreased from an initial 254.0 F g^{-1} to a final 216.4F·g^{-1}, MoO_3-NS from 123.3 F g^{-1} to 100.2 F g^{-1}, and the MoO_3-NB from 8.7 F g^{-1} to 6.9 F g^{-1}, which retain 85.2 %, 81.2 % and 79.1 % of the initial capacitance, respectively. Therefore, the capacitance of MoO_3-NPs showed excellent stability in the 2000 cycles due to the specific morphology with high S_{BET}, suitable V_p size and more accessible active sites on their surfaces, resulting in both moderate nanoprism connection and void channel interconnection [65]. It would be beneficial to have a sufficiently large contact at the electrolyte/electrode interface to achieve a high specific capacitance, which would be favorable for the huge potential of low cost and high performance supercapacitors. However, compared MoO_3 with the similar supercapacitors in previously published literature as shown in Table 1, the highest specific capacitance of MoO_3-NPs was still lower than that of some previous CEs, which was caused by the intrinsic catalytic characteristics with different morphologies, as well as the encapsulated conditions of supercapacitors. Therefore, this experiment could provide a possible way to select a suitable preparation method of MoO_3 as the expected electrode materials for fabrication of supercapacitors.

To understand the performance of different morphologies of MoO_3, symmetrical supercapacitors (SSC) with MoO_3//MoO_3 were assembled by sandwiching of a hydrophilic cellulose separator between two identical electrodes, followed by immersion in 1.0 mol L^{-1} Na_2SO_4 electrolyte. All electrochemical measurements of the symmetrical capacitors were performed in double electrode configuration. Fig. 6(a–c) present the CV profiles of SSC at different scan rate. The range of 0.0 V–1.0 V was selected as the optimal operating voltage of the SSC device [66]. With the increase of scanning rate, the CV curves of MoO_3-NP and MoO_3-NS remain nearly rectangular shapes, indicating that the devices have good electrochemical reversibility. Fig. 6(d–f) present the GCD profiles of SSC measured at different current densities. The appearance of charging/-discharging curves were indicative of pseudocapacitive behavior in accordance with GCD profiles. Fig. 6(g) draws the rate performance of the three molybdenum oxides. On one hand, the specific capacitances of the three oxide electrodes increase with the decrease of loading current density. The increased capacitance values are believed to be a result of the morphology-dependent of the MoO_3 nanoparticles, which can enhance electrolyte ion mobility to the electroactive sites and increase the faradaic reactions. As a result, the Ragone plot of Fig. 6(h) showed that the power density of 11702 W kg^{-1} and energy density of 43.3 Wh·kg^{-1} was demonstrated by the MoO_3-NP at 0.3 A g^{-1}, followed by the performance of the MoO_3-NS, which displayed energy density of 32.6 Wh·kg^{-1}. In contrast, the MoO_3-NB demonstrated the least performance due to the irregular morphology [67]. The Nyquist plots all display a semicircle followed by a sloping straight line in Fig. 6(i).The charge-transfer resistance (R_{ct}) values were 1.29, 2.78 and 6.36 Ω cm^2 of MoO_3-NP, MoO_3-NS and MoO_3-NB electrode, respectively. It can be seen that MoO_3-NP has the smallest semicircle, demonstrating that it has the smallest charge transfer resistance among all electrodes tested, and demonstrates its good electrical conductivity and low resistance. Herein, the MoO_3-NP with the largest S_{BET} and a suitable aspect ratio can generate certain physicochemical structures to provide a greater number of reactivity sites and the uniformly distribution porous structure, which can promote ion adsorption on the electrode surface and the ion diffusion in the electrolyte [68]. In particular, due to a larger number of exposed facets coupled with the orthotropic nanoprism architecture, the MoO_3-NP possess a higher level of vacancy, edge, kinks, terraces and other defect structures, which increase the interface active centers and more powerful charge storage capability. These results confirm that the MoO_3-NP electrode has good electrode performance with high capacitance, rate capability and low R_{ct} in an symmetric supercapacitor device.

4. Conclusions

The different morphology MoO_3 was fabricated using a simple and inexpensive method, and the effect of the morphology-dependent of the

Table 1
Comparisons of the other reported works of MoO_3 electrode.

Materials	Potential window(V)	Electrolyte	Current density	SC(F g^{-1})	Ref.
α-MoO_3 microspheres	−1.0 - 0.2	KCl	0.2 A g^{-1}	201	[45]
MoO_3 nanowires	−0.6 - 0.6	$LiClO_4$	1 A g^{-1}	80	[49]
α-MoO_3 nanobelts	−0.55 - 0.0	H_2SO_4	5 mV s^{-1}	8.8	[50]
MoO_3 nanosheets	−0.6–0.2	LiCl	0.2 A g^{-1}	829	[50]
a-MoO_3 film	−0.5 - 0.0	Na_2SO_4	6 A g^{-1}	1249	[61]
S-MoO_3	0.0–0.9	LiCl	10 mV s^{-1}	220	[62]
MoO_3 nanorods	0.0–1.0	Na_2SO_4	1 mA g^{-1}	176	[66]
MoO_{3-x} microplates	0.6–0.2	H_2SO_4	20 A g^{-1}	410.6	[67]
MoO_{3-x} microdisks	0.6–0.2	H_2SO_4	20 A g^{-1}	226.4	[67]
MoO_3 nanoprisms	−0.2 - 0.5	Na_2SO_4	0.3 A g^{-1}	254.0	Present work
MoO_3 nanostrips	−0.2 - 0.5	Na_2SO_4	0.3 A g^{-1}	123.3	Present work
MoO_3 nanobulks	−0.2 - 0.5	Na_2SO_4	0.3 A g^{-1}	8.7	Present work

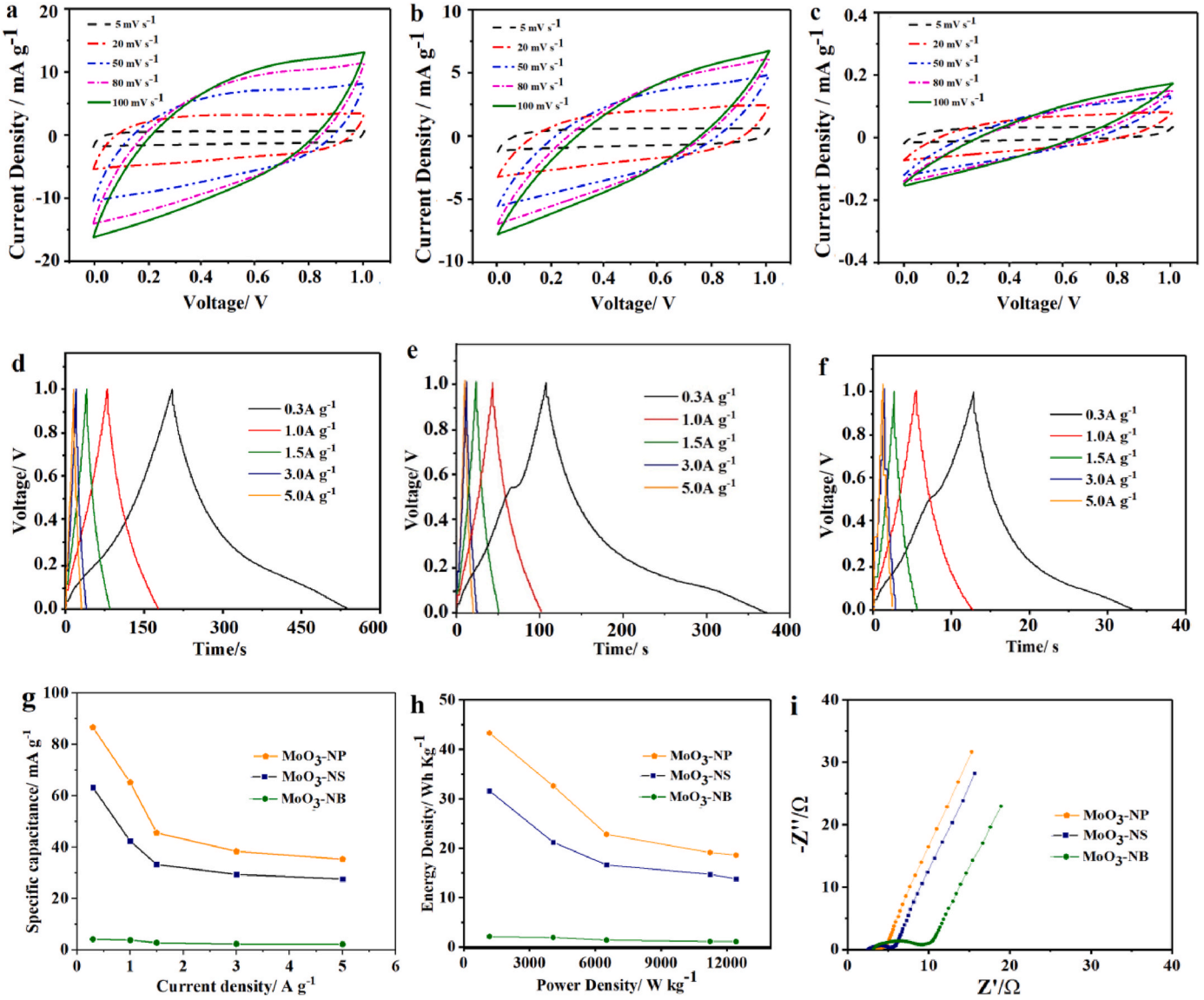


Fig. 6. Electrochemical performance of MoO_3 with different morphology for SSC in 1.0 mol L^{-1} Na_2SO_4. CV profiles of (a) MoO_3-NP, (b) MoO_3-NS (c) MoO_3-NB electrode at different scan rates. GCD curves of (d) MoO_3-NP, (e) MoO_3-NS, (f) MoO_3-NB electrodes at different current density. (g) Nyquist plots and (h) Ragone plots of SSC.

electrode material on the supercapacitor performance was extensively investigated. The galvanostatic charge-discharge were measured in 1 mol L^{-1} Na_2SO_4 electrolyte at a current density of 0.3 A g^{-1} through three consecutive measurements, and the specific capacitances of MoO_3-NP, -NS and -NB electrodes reached 254.0 ± 3.5, 123.3 ± 3.0 and 8.7 ± 1.0 F g^{-1}, respectively. Comparing the morphologies, the MoO_3-NP with the largest S_{BET} and V_p values reduces the formation density of the electrode, which can increase the electrochemical active sites to enhance

reactions and disperse distribution channels to promote ion diffusion, while reducing the interfacial R_{ct}. Furthermore, morphology-dependent leads to an increase in the specific capacity and favors the energy conversion efficiency of the electrochemical process in supercapacitor. The relationships between morphology-dependent and effective activity of different MoO_3 indicate that the MoO_3-NP as electrode achieves an improvement in the specific capacitance of about 2.2 and 29.2 times, respectively, as compared to that of MoO_3-NS and MoO_3-NB as electrode in supercapacitors. In particular, due to the large specific capacitance, outstanding cycle stability, and superior capacitance retention ability, the MoO_3-NP can be applied to promising TMOs electrode materials for the supercapacitor applications.

CRediT authorship contribution statement

Hui Zhao: Writing – original draft, Investigation, Data curation. **Ziwei Chen:** Investigation, Data curation. **Yuanhao Li:** Formal analysis, Data curation. **Shuhui Liu:** Investigation, Formal analysis, Data curation. **Bei Ruan:** Formal analysis, Data curation. **Mingxing Wu:** Project administration, Conceptualization. **Ping Li:** Writing – original draft, Conceptualization. **Kezhong Wu:** Writing – review & editing, Project administration, Methodology, Conceptualization.

Declaration of Competing interest

The authors declare that they have no known competing financial interestssor personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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